

① Codes for signed integers $[-128, 127]$



We can represent only numbers from $[-128, 127]$ on 8 bits.

$$127 = 01111111$$

$$-128 = 10000000$$

$$[-2^{m-1}, 2^{m-1}-1]$$

a) $x = 103(10) \rightarrow ?(2)$

sum of powers of 2

$$x = 64 + 32 + 4 + 2 + 1 = 2^6 + 2^5 + 2^2 + 2^1 + 2^0$$

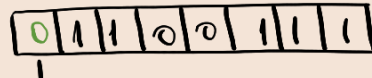
write on 8 bits $\rightarrow 0$ where the index is not there
 $\rightarrow 1$ where the index is there

$$x = \underline{0} \underline{1} \underline{1} \underline{0} \underline{0} \underline{1} \underline{1} \underline{1} = 1100111(2)$$

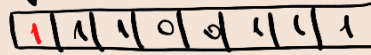
b) $y = 54(10) \rightarrow ?(2)$

$$y = 2^5 + 2^4 + 2^2 + 2^1 = \underline{0} \underline{0} \underline{1} \underline{1} \underline{0} \underline{1} \underline{1} \underline{0} = 110110(2)$$

c) $+ [103]_{dir} = [103]_{inv} = [103]_{comp} =$



$[-103]_{dir} =$



$[-103]_{inv} =$



1's comp. from dir.

or

$$[-103]_{inv} = \begin{array}{r} \boxed{11111111} \\ - \boxed{01100111} \\ \hline \end{array}$$

$[103]_{dir}$

$$\begin{array}{r} \boxed{11111111} \\ - \end{array}$$

$$\boxed{01100111}$$

$$\boxed{11000111}$$

$[-103]_{comp} = \boxed{11000111} \quad 2's \text{ comp. from dir.}$

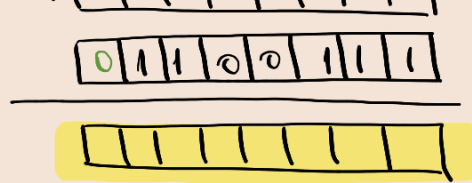
or

$$[-103]_{comp} = \begin{array}{r} 1 \boxed{00000000} \\ - \boxed{01100111} \\ \hline \end{array}$$

$[103]_{dir}$

$$\boxed{10000000}$$

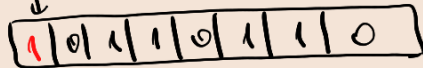
See second branch
in def



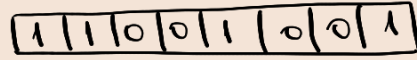
d) $+ [54]_{\text{dir.}} = [54]_{\text{inv.}} = [54]_{\text{comp.}}$



$[-54]_{\text{dir.}}$

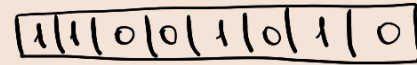


$[-54]_{\text{inv.}}$



1's comp. from dir

$[-54]_{\text{comp.}}$



2's comp from dir.

$$[x+y]_{\text{comp}} = [x]_{\text{comp}} \oplus [y]_{\text{comp}}$$

$$[x-y]_{\text{comp}} = [x]_{\text{comp}} \oplus [-y]_{\text{comp}}$$

e) $[103+54]_{\text{comp}} = [103]_{\text{comp}} + [54]_{\text{comp}} =$

overflow because the operands are "+"
the result is "-".

f) $[103-54]_{\text{comp}} = [103]_{\text{comp}} + [-54]_{\text{comp}} =$

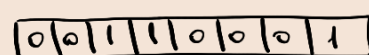
not an overflow because the operands are "+"
=> we eliminate that 1.

back to B10 $\rightarrow 2^0 + 2^4 + 2^5 = 32 + 16 + 1 = [49]_{\text{comp.}}$

g) $[54-103]_{\text{comp}} = [54]_{\text{comp}} + [-103]_{\text{comp}} =$

$[11110010] = [-49]_{\text{comp.}}$

2's comp. $\rightarrow$ we need to find the absolute value first.



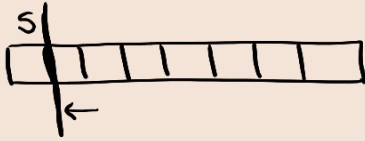
$= 2^0 + 2^4 + 2^5 = 1 + 16 + 32 = 49$

h) $[54-54]_{\text{comp}} = [54]_{\text{comp}} + [-54]_{\text{comp}} =$

1 1 0 0 1 0 1 0

1 0 0 0 0 0 0 0 = [0] comp.
 ↳ the complementary code for 0.

② Codes for signed subunitary numbers $[-1, 1-2^{-7}]$ 8 bits



a) + $x = 0,73$

$$+ y = \frac{10}{16} = 10 \cdot 16^{-1} = 0, A_{(16)} = 0, 1010 (2)$$

$$\downarrow$$

$$A \cdot 16^{-1} = 10 \cdot 16^{-1} = \frac{10}{16}$$

$$\downarrow$$

$$A_{(16)} = 2^3 + 2^1 = 1010 (2)$$

$$x = 0,73 \cdot 8 = 5,84$$

$$0,84 \cdot 8 = 6,72$$

$$0,72 \cdot 8 = 5,76$$

$$0,76 \cdot 8 = 6,08$$

$$x = 0,565 (2) = 0, 101, 110, 101 (2)$$

$2^3 \Rightarrow$ groups of 3

$x: \begin{array}{|c|c|c|c|c|c|c|c|} \hline 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ \hline \end{array}$

$y: \begin{array}{|c|c|c|c|c|c|c|c|} \hline 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ \hline \end{array}$

$$y: \left[\frac{10}{16} \right]_{dir} = \left[\frac{10}{16} \right]_{inv} = \left[\frac{10}{16} \right]_{comp} = \begin{array}{|c|c|c|c|c|c|c|c|} \hline 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ \hline \end{array}$$

$$\left[-\frac{10}{16} \right]_{dir} = \begin{array}{|c|c|c|c|c|c|c|c|} \hline 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ \hline \end{array}$$

$$\left[-\frac{10}{16} \right]_{inv} = \begin{array}{|c|c|c|c|c|c|c|c|} \hline 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 \\ \hline \end{array}$$

$$\left[-\frac{10}{16} \right]_{comp} = \begin{array}{|c|c|c|c|c|c|c|c|} \hline 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ \hline \end{array}$$

$$x: [0,73]_{dir} = [0,73]_{inv} = [0,73]_{comp} = \begin{array}{|c|c|c|c|c|c|c|c|} \hline 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ \hline \end{array}$$

$$\left[-0,73 \right]_{dir} = \begin{array}{|c|c|c|c|c|c|c|c|} \hline 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ \hline \end{array}$$

000.
 001.
 010.
 011.
 100.
 101. → 5
 110. → 6

$$[-0,73]_{inv} = \boxed{1 \mid 0 \mid 1 \mid 0 \mid 0 \mid 0 \mid 1 \mid 0}$$

$$[-0,73]_{comp} = \boxed{1 \mid 0 \mid 1 \mid 0 \mid 0 \mid 0 \mid 1 \mid 1}$$

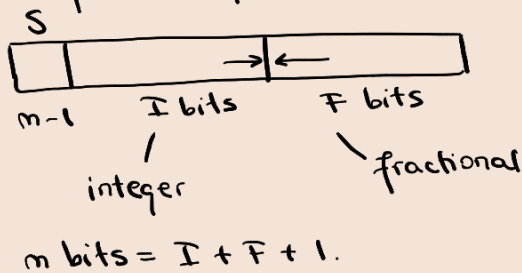
$$b) \left[\frac{10}{16} + 0,73 \right]_{comp} = \left[\frac{10}{16} \right]_{comp} \oplus [0,73]_{comp} = \begin{array}{r} \boxed{0 \mid 1 \mid 0 \mid 1 \mid 0 \mid 0 \mid 0 \mid 0} \\ + \boxed{0 \mid 1 \mid 0 \mid 1 \mid 1 \mid 1 \mid 0 \mid 1} \\ \hline \boxed{1 \mid 0 \mid 1 \mid 0 \mid 1 \mid 1 \mid 0 \mid 1} \end{array}$$

↓
overflow because two "+" ms. added cannot be "-".

$$c) \left[\frac{10}{16} - 0,73 \right]_{comp} = \left[\frac{10}{16} \right]_{comp} \oplus [-0,73]_{comp} = \begin{array}{r} \boxed{0 \mid 1 \mid 0 \mid 1 \mid 0 \mid 0 \mid 0 \mid 0} \\ + \boxed{1 \mid 0 \mid 1 \mid 0 \mid 0 \mid 0 \mid 1 \mid 1} \\ \hline \boxed{1 \mid 1 \mid 1 \mid 1 \mid 1 \mid 0 \mid 1 \mid 1} = [-0,101] \end{array}$$

2's → abs. value: $\boxed{0 \mid 0 \mid 0 \mid 0 \mid 1 \mid 1 \mid 0 \mid 1} = [0,101]$
 $= 2^{-4} + 2^{-5} + 2^{-7} = 0,101$

③ Fixed-point representation



a) m = 32 bits

I = 13

F = 18

x = 2050,53

2050 = 2048 + 2 = $2^8 + 2^1$ = $\overbrace{00001000000010}^{I=13 \text{ bits}}$ = 100000010 (2)

0,53 = ? (8) = ? (2)



$2^3 \Rightarrow F: 3$ multiplications by 8.

0,53 · 8 = 4,24

0,24 · 8 = 1,92

0,92 · 8 = 7,36

0,36 · 8 = 2,88

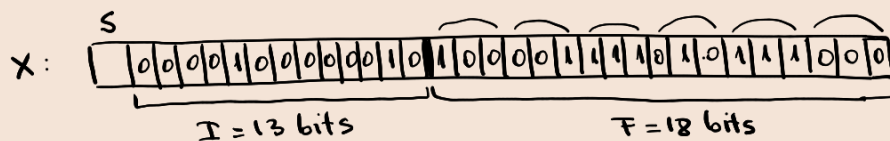
0,88 · 8 = 7,04

0,04 · 8 = 0,32

$\Rightarrow 0,53_{(10)} = 0,417270_{(8)}$ $2^3 \Rightarrow$ groups of 3.
 $= \boxed{0,10000111,010111000}_{(2)}$

000
001
010
011
100 → 4
101
110
111

$$\Rightarrow X = 100000010, 100\ 001\ 111\ 010\ 111\ 000\ (2) = 2050,53\ (10).$$



"Exercises propositional logic" - HW:

EX 8.4: Write all the anti-models of the formula using CNF.

CNF = conjunction of disjunctions $(\vee) \wedge (\vee)$

↳ distribute $\vee$ over $\wedge$

$$U_4 = p \vee \neg(q \wedge \neg r) \rightarrow p \wedge q \wedge \neg r$$

1. replace the " $\rightarrow$ ":

$$\neg(p \vee \neg(q \wedge \neg r)) \vee (p \wedge q \wedge \neg r)$$

2. De Morgan:

$$(\neg p \wedge (q \wedge \neg r)) \vee (p \wedge q \wedge \neg r)$$

$$\underbrace{(\neg p \wedge q \wedge \neg r)}_B \vee \underbrace{(p \wedge q \wedge \neg r)}_A$$

3. Distribution:

$$(B \wedge A) \vee (C \wedge A) \equiv$$

$$(A \wedge B) \vee (A \wedge C) \equiv A \wedge (B \vee C)$$

$$(q \wedge \neg r) \wedge (\underbrace{\neg p \vee p}_{\text{tautology}})$$

$$\Rightarrow U_4 = q \wedge \neg r$$

! U_4 is false $\Leftrightarrow$ CNF is false $\Leftrightarrow$ at least one conjunct is false.

$\Rightarrow U_4$ is false when $q = F$ or $r = T$.

p	q	r	$\neg r$	U_4
T	T	T	F	F
T	T	F	T	T
T	F	T	F	F

T	F	F	T	F
F	T	T	F	F
F	T	F	T	T
F	F	T	F	F
F	F	F	T	F

$\Rightarrow$ 2 models
 $\Rightarrow 8-2 = 6$ antimodels.